RIDMIKA YOHANI EKANAYAKA

Undergraduate

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LinkedIn — GitHub — HackerRank — Portfolio

PERSONAL SUMMARY

Final year Software Engineering undergraduate at Sri Lanka Technology Campus with a strong interest in Quality Assurance and software testing. Familiar with manual testing concepts, test case design, functional and regression testing and defect reporting. Possess a foundation in software development, databases and API testing, with experience developing web and full-stack applications through academic and personal projects. Detail-oriented, analytical and eager to apply testing knowledge, improve software quality and gain practical industry experience through a QA internship.

EDUCATIONAL QUALIFICATIONS

BSc (Hons) in Software Engineering, Sri Lanka Technology Campus, Padukka — 2023 – Present

G.C.E. Advanced Level (Physical Science), Ferguson High School, Ratnapura — 2021

G.C.E. Ordinary Level, Ferguson High School, Ratnapura — 2018

- Results: A – 5, B – 2, C – 2

TECHNICAL SKILLS

Quality Assurance: Manual Testing, Functional Testing, Regression Testing, Smoke Testing, Sanity Testing, UI Testing, Exploratory Testing

Test Design & Documentation: Test Case Design, Test Scenario Creation, Test Planning, Requirement Analysis, Bug/Defect Reporting

API Testing: Postman, REST API Testing Concepts, HTTP Methods, Status Codes, JSON

Automation Testing: Selenium, Python (Basic)

Programming Languages: Python, Java, JavaScript, C, C++, HTML, CSS, SCSS

Frameworks & Libraries: React, Node.js, Express.js, Flutter, Bootstrap

Databases: MySQL, MongoDB

Tools & Collaboration: Git, GitHub, VS Code, Overleaf

Software Development Methodologies: SDLC, STLC, Agile/Scrum Fundamentals

PROJECTS

Human-Centered Explainable AI Framework for Loan Approval — Final Year Project, Ongoing (2026)

AI-driven loan approval systems often act as "black boxes," making decisions without explaining the reasoning behind them and sometimes treating applicant groups unfairly. This ongoing project proposes an XAI-based framework using tree-based machine learning models and SHAP explanations to make loan approval decisions more transparent, along with fairness metrics to help reduce bias. The system is being developed with a Flutter frontend and a Node.js/Express API backend.

Technologies: Python, Machine Learning, SHAP, Fairness Metrics, Flutter, Node.js, Express.js

SmartCare – Patient Readmission Prediction | Link — Academic Project, September 2026

Built a machine learning pipeline to predict 30-day hospital patient readmissions using Random Forest, Decision Tree, XGBoost, and KNN models on a 1,000-record clinical dataset. Applied SHAP to provide model explainability and deployed a Streamlit prototype for clinical decision support. Achieved a ROC-AUC score of 0.942 and 100% recall using the Random Forest model.

Technologies: Python, Scikit-learn, XGBoost, SHAP, Pandas, Streamlit

OGBN-Arxiv GNN Node Classification | Link Academic Project, August 2026

Implemented and compared Graph Convolutional Network (GCN) and Graph Attention Network (GAT) models for node classification on a 169,343-node citation dataset. Achieved 54.6% test accuracy, analyzed model behavior through embedding visualizations and attention-weight analysis, and presented findings using an interactive Streamlit dashboard.

Technologies: Python, PyTorch, PyTorch Geometric, Scikit-learn, Streamlit, Google Colab

TrendVerse – Fashion E-Commerce Website | Link Personal Project, February 2026

Developed a responsive full-stack fashion e-commerce website focused on providing a smooth and user-friendly shopping experience. Implemented features such as product browsing, search and filtering, cart management, checkout, user dashboards, and admin controls. The project provides a practical environment for validating user workflows, checking input handling, testing API functionality, and verifying application behavior across different scenarios.

Technologies: React, JavaScript, HTML, SCSS, Node.js, Express.js, MongoDB

Utopia – Digital Citizen & Government Services Platform | Link Academic Project, July 2025

Utopia is a digital citizen and government services platform designed to simplify how citizens interact with government processes online, reducing reliance on manual paperwork and in-person visits. The backend service was implemented using Node.js and Express, with MongoDB (via Mongoose) used for structured data storage of citizen and government records. JWT (JSON Web Tokens) was used to implement secure authentication and protect user sessions, Multer was used to handle file uploads such as documents and identification records, and Nodemailer was integrated to support automated email notifications, such as confirmations or status updates, as part of the citizen-facing workflow.

Technologies: Node.js, Express.js, MongoDB, Mongoose, JWT, Multer, Nodemailer

CERTIFICATIONS

Power BI for Beginners – Simplilearn

Quality Assurance Courses with Certificate – Simplilearn

Full Stack Development Certificate Course, AIO-IT SOLUTIONS (PVT) LTD

Full Stack Development Certificate Course, SKYRE

Python (Basic) Certificate – HackerRank

Power BI Data Modeling with DAX, Simplilearn

VOLUNTEERING & EXTRA-CURRICULAR ACTIVITIES

Treasurer, Computer Society Student Branch Chapter of SLTC IEEE 2025 – 2026

- Managed tentative budgets and final financial reports for student events.
- Supported activities including Codemania v6.0, GitGenius 2026, and CloudSpace v1.0.

IEEE Member & Volunteer, SLTC IEEE 2024–2026

- Logistics Sub-Committee Deputy Head, IEEE CS Student Branch Chapter, SLTC — 2024–2025
- Promotion Team Head, IEEE Codemania V5.0 — IEEE CS Student Branch Chapter, SLTC
- Coordination Team Deputy Head, IEEE Day 2025 — IEEE CS Student Branch Chapter, SLTC
- Program Team Deputy Head, Arduino Challenge 2025 — IEEE CS Student Branch Chapter, SLTC
- Program Team Member, Decode Xtreme 2024 / Git Genius 2024 — IEEE CS Student Branch Chapter, SLTC
- Volunteered at Codemania v5.0, GitGenius 2024, and DecodeXtream.
- Served in roles including Head, Deputy Head, and Member for various event activities.